

Duy Nguyen

dnguyen44@seattleu.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Seattle, WA

EDUCATION

Master of Science in Data Science

September 2025 – Expected June 2027

Seattle University

Seattle, WA

- **Relevant Coursework:** Deep Learning (transfer learning, CNN architectures, model optimization), Statistical Inference & Experimental Design (resampling methods, correlation/regression, chi-square tests, confidence intervals), Programming for Data Science (Python, algorithms), Foundations of Data Science (calculus/linear algebra optimization, image classification, data manipulation, EDA, regression with ANOVA F-tests)
- **Current GPA:** 4.0

Professional Certificate in Machine Learning and Artificial Intelligence

July 2024

UC Berkeley College of Engineering and Haas School of Business

Online Bootcamp

- Built predictive models using ensemble methods (Random Forest, XGBoost) and deep learning (LSTM, custom MLP) to forecast hospital operations, achieving 80% accuracy through systematic feature engineering and hyperparameter optimization
- Developed end-to-end ML pipeline including data cleaning, feature extraction, model validation, and business impact analysis projecting \$30.4M annual savings
- Collaborated with 150+ technical peers through code reviews and presentations, with methodology selected as program exemplar (LinkedIn post)

Bachelor of Arts in Economics, Data Analysis concentration

Dec 2022

Simon Fraser University

Burnaby, BC

- Applied econometric methods and causal inference techniques (Difference-in-Differences, Regression Discontinuity Design) to analyze policy impacts using Python, producing research-quality analysis recognized by faculty (LinkedIn recommendation)

TECHNICAL SKILLS

Programming Languages: Python, R, LaTeX

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, FastAI, Transfer Learning, CNNs, Ensemble Methods (Random Forest, XGBoost), Hyperparameter Tuning, Model Optimization

Data Science: pandas, NumPy, Feature Engineering, Data Cleaning, Model Validation, Cross-Validation

Statistical Analysis: Regression Analysis, ANOVA, Hypothesis Testing, Experimental Design, A/B Testing, Causal Inference

Data Visualization: Matplotlib, Seaborn, Plotly, Tableau (exposure)

Tools & Frameworks: Git/GitHub, Jupyter Notebook/Lab, VS Code, Neo4j, LlamaIndex, Langchain

PROJECTS

AI-Powered Portfolio Website with Intelligent Assistant | *Personal Project*

Present

- Engineered full-stack portfolio website from scratch using Flask backend and responsive frontend, integrating Claude API-powered conversational AI to help recruiters efficiently learn about candidate qualifications and technical expertise
- Implemented RESTful API architecture with CORS configuration, conversation memory management, and separate evaluation endpoints, demonstrating software engineering best practices including clean code organization and production deployment on Render
- Built dual-chatbot system using LangChain framework: (1) portfolio assistant for recruiter queries with role specific responses, and (2) algorithm evaluation
- Designed self-healing error correction system using tertiary AI fallback mechanism to detect and automatically reformat malformed responses, ensuring robust user experience through multi-layered validation and error handling
- Developed comprehensive formatting system translating markdown responses to styled HTML with animated typing indicators, drag-and-drop positioning, and dynamic resizing for optimal user experience
- Technologies: Python, Flask, JavaScript, HTML/CSS, LangChain, Claude API (Anthropic), Neo4j, Git/GitHub

Aircraft Fuel Efficiency Statistical Analysis | *Seattle University DATA 5100*

Dec 2025

- Analyzed 1.88 million flight measurements using Python (pandas, NumPy) and SQL to quantify operational efficiency factors, managing comprehensive data pipeline from extraction through validation
- Built predictive regression models achieving 95.9% accuracy through systematic ANOVA, nested F-tests, and variance decomposition, identifying that engine performance accounted for 64% of explainable variance
- Developed interactive visualizations using Matplotlib, Seaborn, and Plotly to communicate complex statistical findings, translating technical results into strategic business recommendations that challenged conventional operational priorities
- Documented comprehensive analytical methodology including diagnostic procedures and model assumptions, ensuring reproducible research standards

Automated Garbage Classification System | *Seattle University DATA 5100*

Dec 2025

- Developed production-ready computer vision system using transfer learning with ResNet34 to classify 2,527 waste images into 6 categories, achieving 94% accuracy with 100% minority class recall through systematic experimentation
- Designed and executed controlled experiments testing 4 different class imbalance mitigation strategies (oversampling, weighted loss, augmentation variations), discovering that conservative augmentation (10° rotation, 1.1x zoom) outperformed aggressive approaches by 0.5% while being faster and more maintainable
- Applied rigorous experimental methodology including confusion matrix analysis, per-class performance evaluation, and hyperparameter optimization across 5-epoch training cycles, identifying critical "double-weighting" problem when combining techniques
- Validated real-world generalization through external image testing, demonstrating 99.9% confidence classification on unseen data with complex backgrounds, positioning model for recycling facility deployment with projected 60% labor cost reduction

Marketing Campaign Optimization | *UC Berkeley ML/AI Certificate*

Jun 2024

- Conducted exploratory data analysis using Python to identify customer subscription drivers, discovering correlation between education level and subscription rates through statistical validation
- Built classification model using Support Vector Machines to optimize marketing resource allocation, identifying optimal customer contact windows (6-8 minute calls) through systematic feature importance analysis
- Translated technical findings into strategic business recommendations for marketing investment optimization, demonstrating ability to communicate insights to non-technical stakeholders

Medical Image Classification | *Kaggle Competition*

Nov 2023

- Fine-tuned ResNet50 convolutional neural network using PyTorch for medical image classification, achieving 57.1% accuracy and ranking 12th place on competition leaderboard through systematic transfer learning and model optimization

EXPERIENCE

AI Engineer

Mar 2024 – Oct 2024

Simon Fraser University Blueprint

- Developed AI chatbot serving 660K+ user interactions for MOSAIC immigrant services, implementing Neo4j graph database architecture and multi-language natural language processing to improve service accessibility
- Achieved 90% response accuracy by embedding domain knowledge and implementing automated quality validation through secondary evaluation system
- Collaborated with cross-functional team of engineers to identify and resolve frontend rendering issues, improving user experience through iterative problem-solving and technical communication
- Applied pattern detection methodologies to large user interaction datasets, implementing fallback mechanisms that improved system reliability, shortlisted Top 4 for SFU CS Diversity Award

Volunteer Software Developer

Jan 2024 – May 2024

Simon Fraser University Faisal Lab

- Developed retrieval-augmented generation (RAG) pipeline using Python and LlamaIndex framework to automate medical documentation workflow, eliminating manual report lookup processes
- Implemented iterative testing and validation pipeline through systematic unit testing, ensuring deterministic JSON parameter generation from natural language physician inputs
- Created comprehensive documentation for chatbot implementation, enabling physicians to efficiently retrieve CT/MRI scan parameters and accelerating report generation workflow by 50%